

CMS Claims and Provider Data Analysis

Capstone Project

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ABSTRACT

The U.S. healthcare system's shift from fee-for-service to **value-based care (VBC)** models, including Accountable Care Organizations (ACOs) and bundled payments, aims to improve patient outcomes and optimize spending. However, the true financial and clinical impact, especially concerning persistent healthcare disparities based on race, ethnicity, and socioeconomic status, remains under investigation. This capstone project addresses these critical gaps by analyzing large-scale claims and premium data to evaluate the **cost-effectiveness, care outcomes, and equity implications** of VBC models.

Utilizing advanced statistical and machine learning techniques with tools like SAS Studio, R, Python (Jupyter Notebook, Pandas), and PostgreSQL, the study explores several key areas. We found that provider availability significantly impacts service utilization, with higher provider density (e.g., Michigan, Arizona) correlating with broader patient coverage, while rural areas often show lower access. Analysis of **Managed Care Organization (MCO) efficiency** revealed that Comprehensive MCOs serve the largest patient populations (11,211 active patients) across diverse services like behavioral health and prenatal care, indicating broad integrated delivery. Trends in active patients and services per patient across states highlighted Michigan's robust care intensity growth (from 24,921 to 38,301 services per patient, 2020-2024), contrasting with Nevada's significantly lower utilization.

A critical finding concerns **discrepancies between average Medicare payments and submitted charges** across Diagnosis-Related Groups (DRGs) and regions. Submitted charges consistently exceeded Medicare payments, often by a 4-5x margin (e.g., "Cardio and Pulmonary Related" DRGs: ~\$99.9M paid vs. ~\$542.8M submitted). High-volume DRGs like "Other complications" and "Cardio and Pulmonary Related" also represented substantial financial footprints. Geographically, urban areas accounted for considerably higher Medicare payments (\$2.49B) and submitted charges compared to rural areas (~\$6.2M), highlighting stark regional disparities in service volume and financial flows. These insights provide actionable information for healthcare stakeholders to refine VBC strategies, promoting both financial sustainability and improved, equitable patient outcomes, particularly for high-risk and underserved populations.

Keywords: value-based care, healthcare disparities, claims data, cost-effectiveness, patient outcomes, health equity, Medicare, managed care organizations, DRG, provider access

INTRODUCTION

The transition from fee-for-service to value-based care models represents a fundamental shift in the U.S. healthcare system, aiming to enhance patient outcomes while optimizing healthcare spending. Models such as Accountable Care Organizations (ACOs) and bundled payment systems have been introduced to promote more coordinated, efficient, and patient-centered care. However, the true financial and clinical impact of these models, particularly as reflected in claims data and member premium costs, remains an area of ongoing investigation.

At the same time, persistent healthcare disparities based on race, ethnicity, and socioeconomic status continue to challenge the goal of equitable care. While value-based care programs are designed to improve access and quality across all populations, there is a risk that misaligned financial incentives could unintentionally widen existing disparities rather than close them.

This research addresses these critical gaps by analyzing large-scale claims and premium data to evaluate the cost-effectiveness, care outcomes, and equity implications of value-based payment models. It further explores patterns of provider access, care intensity, utilization trends, and regional payment disparities. By leveraging advanced statistical methods and machine learning techniques, the study aims to generate actionable insights for healthcare providers, payers, and policymakers seeking to refine value-based care strategies that deliver both financial sustainability and improved patient outcomes, especially for high-risk and underserved populations.

OBJECTIVES

- **Access to Care:** Evaluate the relationship between the number of providers and the percentage of eligible patients receiving services across counties and states, identifying regions with potential access challenges.
- **Care Intensity and Efficiency:** Identify which MCOs or plan categories deliver the highest number of services per active patient to assess efficiency and inform value-based contracting strategies.
- **Utilization Trends:** Analyze trends in the number of active patients and services per patient across state overtime to detect shifts related to policy, population changes, or program performance.
- **Cost Efficiency and Payment Disparities:** Compare average Medicare payments with submitted charges across diagnosis-related groups (DRGs) and geographic areas to highlight discrepancies in reimbursement practices.
- **High-Impact Conditions:** Identify which DRGs generate the highest discharge volumes and analyze how associated payments and charges vary by state to inform strategic resource planning.
- **Health Disparities and Risk Profiles:** Assess how average patient risk scores and the prevalence of chronic conditions vary by demographics (age, race, dual eligibility) and geography (state, urban vs. rural), identifying vulnerable populations and care gaps.
- **Urban vs. Rural Payment Patterns:** Evaluate differences in Medicare payment and charge patterns between urban and rural providers to identify geographic inequities in healthcare financing.

STUDY OVERVIEW

This study investigates the effectiveness, efficiency, and equity of value-based care models by analyzing Medicaid and Medicare administrative datasets. Guided by a set of targeted research questions, the analysis focuses on access to care, service utilization, payment discrepancies, and health disparities across geographic and demographic dimensions. Key questions explore how provider availability influences service uptake, how Managed Care Organizations (MCOs) vary in care delivery intensity, and how Medicare payment patterns differ by diagnosis, region, and provider type.

Hypotheses are developed to test assumptions around cost effectiveness, care coordination, and risk-adjusted outcomes in value-based environments. The literature review highlights gaps in existing research, particularly regarding the real-world financial and clinical impacts of models like ACOs and bundled payments, as well as their implications for health equity.

The study utilizes de-identified data from multiple national datasets, including Medicaid Managed Care metrics and Medicare inpatient hospital records. Analytical methods include regression modeling, trend analysis, propensity score matching, and survival analysis. These approaches are chosen to identify causal relationships, control for confounding factors, and generate actionable insights for improving the design and implementation of value-based care strategies.

RESEARCH QUESTIONS

Using the four selected datasets, I can develop several important research questions that may benefit patients, providers, payers, and government healthcare agencies by helping to reduce costs and improve patient care. Below are some of the key research questions, along with the corresponding datasets and variables to be analyzed.

Medicaid Managed Care

Sl.No	Research Questions	What problem does it solve?	Variables
1	How does the number of providers affect the percentage of eligible patients receiving services across different counties and states?	This question explores access to care. It can help identify whether areas with more providers are more successful in serving their eligible populations.	Number of Providers Percent of Eligible Patients Receiving Services County, State
2	Which MCOs or plan categories deliver the highest number of services per active patient, and what does this suggest about care intensity or efficiency?	This evaluates service delivery efficiency and MCO performance, helping inform contracting strategies.	Number of Services per Active Patient MCO Name Plan Category Service Category
3	What are the trends in the number of active patients and the number of services per active patient across states?	This gives insight into changes in utilization patterns and reflects policy changes, population shifts, or program effectiveness.	Calendar Year Number of Active Patients Number of Services per Active Patient State

Medicare IP Hospitals by Geography and Service

Sl.No	Research Questions	What problem does it solve?	Variables
4	How do average Medicare payments compare to average submitted charges across different DRGs and geographic regions?	This question assesses cost efficiency and payment discrepancies in the healthcare system. It can reveal whether certain diagnosis-related groups (DRGs) or regions are consistently over- or under-reimbursed relative to charges.	Avg_Mdcr_Pymt_Amt Avg_Submtd_Cvrd_Chrg DRG_Desc / DRG_Cd Rndrng_Privr_Geo_Cd or Rndrng_Privr_Geo_Desc
5	Which DRGs have the highest total discharges, and what are the corresponding average payment amounts and submitted charges?	This evaluates utilization patterns and identifies high-impact conditions (by discharge volume) that are useful for cost and care management strategies.	Tot_Dschrgs DRG_Desc / DRG_Cd Avg_Mdcr_Pymt_Amt Avg_Submtd_Cvrd_Chrg Avg_Tot_Pymt_Amt

Medicare IP Hospitals by Provider and Service

Sl.No	Research Questions	What problem does it solve?	Variables
6	How do average Medicare payments compare to submitted charges across different DRGs and provider locations (urban vs. rural)?	This question examines cost efficiency and geographic payment disparities, helping to identify whether providers in certain areas are more likely to submit higher charges but receive lower payments.	Avg_Mdcr_Pymt_Amt Avg_Submtd_Cvrd_Chrg DRG_Cd / DRG_Desc Rndrng_Privr_RUCA, Rndrng_Privr_RUCA_Desc (rural vs. urban classification) Rndrng_Privr_St or Rndrng_Privr_State_Abrvtn

THE HYPOTHESES

Hypotheses are testable statements used in research to evaluate relationships or differences between variables. The **null hypothesis (H₀)** assumes no effect or difference, serving as the default or baseline. The **alternative hypothesis (H₁)** suggests there is a meaningful effect or difference.

By testing hypotheses, researchers can validate findings and provide actionable insights for patients, providers, payers, or government agencies. The top 4 hypotheses that can be formulated using our dataset are as follows.

Hypothesis 1 - Services per Patient by Plan Category

- **Data set used:** Medicaid_Managed_Care_Q4_2024
- **Hypothesis:** Evaluate if certain plan types (e.g., HMOs vs. PPOs) deliver more services per patient, which may reflect care intensity or access.
 - **H₀:** There is no significant difference in the mean number of services per active patient across different plan categories.
 - **H₁:** There is a significant difference in the number of services per active patient across different plan categories.

Hypothesis 2 - Medicare Payments vs. Charges

- **Data set used:** Medicare_IP_Hospitals_by_Geography_and_Service_2023
- **Hypothesis:** Assess discrepancies between billed and reimbursed amounts to identify inefficiencies or payment gaps.
 - **H₀:** The average Medicare payment amount is equal to the average submitted covered charge for all DRGs.
 - **H₁:** The average Medicare payment amount is significantly different from the average submitted covered charge for at least one DRG.

Hypothesis 3 – Impact of Plan Categories

- **Data set used:** Medicare_IP_Hospitals_by_Provider_and_Service_2023
- **Hypothesis:** MCOs or plan categories that deliver a higher number of services per active patient are associated with increased care intensity, but not necessarily higher care efficiency.
 - **H₀:** There is no significant difference in the number of services per active patient across MCOs or plan categories.
 - **H₁:** There is a significant difference in the number of services per active patient across MCOs or plan categories.

Hypothesis 4 - Discharges by Urban vs. Rural Providers

- Data set used: Medicare_IP_Hospitals_by_Provider_and_Service_2023
- **Hypothesis:** Understand care access and service utilization in different geographic settings.
 - **H₀:** There is no significant difference in mean discharges between rural and urban providers for high-volume DRGs.
 - **H₁:** There is a significant difference in mean discharges between rural and urban providers for high-volume DRGs.

LITERATURE REVIEW

Quality of Primary Care and Cost Implications in Health Systems:

Several studies have examined whether health system affiliation improves primary care quality. Zhou et al. (2020) analyzed data from four U.S. states over five years and found that system-affiliated primary care providers did not consistently deliver higher-quality care compared to independent providers. Metrics such as preventive care delivery, chronic disease management, and patient experience showed marginal differences. This aligns with previous findings by Fisher et al. (2017), who observed that health system consolidation did not uniformly lead to better clinical outcomes or higher patient satisfaction.

Other studies have highlighted the potential benefits of system affiliation, such as enhanced access to resources, shared electronic health records, and care coordination programs (Basu et al., 2018). The question of cost efficiency is central to the discussion of health system integration. Zhou et al. (2020)

reported that system-affiliated providers were associated with higher overall spending, driven primarily by increased service intensity and referrals to specialists within the system.

Factors Influencing Quality and Cost Outcomes:

Several factors complicate the relationship between health system affiliation, quality, and cost. Provider practice patterns, local market competition, and patient population characteristics all influence outcomes. Additionally, the heterogeneity of health systems, ranging from tightly integrated delivery networks to loosely affiliated groups, means that the impact of affiliation can vary widely (Gaynor & Town, 2012).

Impact of MLR:

The study on MLR demonstrates that escalating medical claims, rather than administrative inefficiencies, are the primary drivers of premium increases. This challenges common perceptions and underscores the need for policy interventions focused on controlling underlying medical spending. While the MLR regulation ensures a sizable portion of premiums is spent on care and patient services, it does not limit overall spending; if premiums rise with claim costs, administrative ratios remain compliant. This analysis aligns with findings from the Kaiser Family Foundation and NAIC, confirming that large group plans must dedicate at least 85% of premiums to medical claims. It also complements work exploring insurer market dynamics and rebate distributions in various insurance markets.

Geographic variations:

The article “Geographic Variation in Hospital Costs, Payments, and Length of Stay for Opioid-Related Hospital Visits in the USA” by Mallow et al. (2018) examines regional disparities in hospital costs, payments, and length of stay (LOS) for opioid-related visits across nine U.S. census regions. Using a large dataset from the Vizient Health System Database (2014–2017), the study identified over 193,000 opioid-related inpatient and outpatient visits. The results revealed significant variation: inpatient costs ranged from \$2,894 to \$5,835, and LOS from 3.8 to 5.7 days depending on the region. Outpatient visits showed lower but still variable costs and payments. These disparities highlight inefficiencies and inequities in opioid care delivery. The study suggests that understanding these geographic differences can help policymakers tailor interventions, guide hospital benchmarking, and support more equitable care. While comprehensive, the study is limited by reliance on one hospital network and lacks outcome measures beyond cost and LOS.

RESEARCH DESIGN

Methodology

This study employs a **quantitative** research methodology to evaluate the financial, clinical, and equity impacts of value-based care models, specifically focusing on Medicaid Managed Care and Medicare inpatient data. The approach is structured to provide empirical evidence of the effectiveness and efficiency of these models.

I. Research Design

The research adopts a non-experimental, observational design, utilizing secondary data analysis to explore existing patterns and relationships within the datasets. This design is appropriate for assessing real-world outcomes without manipulating variables, aligning with the study's objectives to understand the current state of value-based care implementation.

II. Data Collection

Data for this study are sourced from publicly available, de-identified datasets:

- **Medicaid Managed Care:** This dataset provides insights into the Managed Care Organization (MCO) performance at the state and county levels. It includes MCO names, service categories, patient counts (active and eligible), provider numbers, service utilization metrics, and plan types over calendar years. It helps analyze care access, provider capacity, and service delivery efficiency within different MCO plans and regions.
- **Medicare Inpatient Hospitals by Geography and Service:** This dataset summarizes healthcare service data by geographic levels, including location codes and descriptions. It tracks Diagnosis-Related Groups (DRGs) with total discharges, average submitted covered charges, average payments from Medicare, and total payers. This enables regional analysis of healthcare utilization, costs, and payment patterns by specific medical conditions or procedures.
- **Medicare IP Hospitals by Provider and Service:** This dataset contains provider and healthcare service details, including numeric IDs (e.g., provider CCN, state FIPS, DRG code), location info (city, state, ZIP), rural-urban classification (RUCA), service descriptors, and utilization metrics like total discharges and average charges/payments, mixing numeric and character data types. These datasets offer comprehensive insights into service delivery patterns, payment structures, and demographic variations, facilitating a robust analysis of value-based care models.

III. Variables

Key variables for analysis include:

- *Independent Variables:* Number of providers, Managed Care Organization (MCO) type, plan category, urban vs. rural classification.
- *Dependent Variables:* Percentage of eligible patients receiving services, number of services per active patient, average Medicare payments, average submitted charges.
- *Control Variables:* Demographic factors such as age, race, dual eligibility status, and geographic location.

These variables are selected to assess the impact of provider availability, plan characteristics, and demographic factors on healthcare outcomes and costs.

Methods and Tools Used

Data analysis is conducted using advanced statistical techniques:

- **Descriptive Statistics:** To summarize and describe the features of the datasets.
- **Regression Analysis:** To identify relationships between independent and dependent variables.
- **Propensity Score Matching:** To control difficult variables and estimate treatment effects.

These methods enable a comprehensive evaluation of the research questions and hypotheses. The following multi-tool approach ensures a robust, flexible, and scalable workflow throughout the capstone project. Some of them are as follows.

- **SAS Studio:** I will use SAS Studio along with R programming for conducting both statistical and prescriptive analyses. These tools are well-suited for handling large datasets and performing complex healthcare data modeling and inference.
- **Jupyter Notebook:** Python will be used extensively in Jupyter Notebook for data cleaning, manipulation, and exploratory data analysis. Python's rich libraries like Pandas, NumPy, and Matplotlib will support efficient processing and visualization of the data.
- **PostgreSQL:** If the project requires managing large volumes of structured data or performing complex queries, PostgreSQL will be used as the database management system to efficiently store and retrieve data.
- **Microsoft Excel:** This is used for data cleansing and creating pivots to get meaningful insights

Limitations

While this study leverages robust Medicaid and Medicare datasets to evaluate value-based care models, several limitations must be acknowledged:

- **Limited Granularity of Claims Data:** Claims data are primarily designed for billing purposes rather than clinical research. As a result, they may lack detailed information on patient health status, severity of illness, social determinants of health, and quality of care indicators, factors that are crucial for fully understanding clinical outcomes and care effectiveness.
- **Absence of Patient-Level Clinical Outcomes:** The datasets do not include direct clinical outcomes (e.g., lab results, functional health scores, patient-reported outcomes), which limits the ability to assess the actual health improvements attributable to value-based care models. Proxy indicators like service utilization or hospital readmissions may not fully capture patient well-being.
- **Potential for Coding and Reporting Bias:** Variability in how providers document services and diagnoses can affect the accuracy of measures such as DRG categorization and average charges. These inconsistencies may distort comparisons across providers or regions.

- **Geographic and Demographic Aggregation:** Due to data privacy protections, some datasets are aggregated at the county or state level. This limits precision in analyzing disparities within subpopulations or identifying high-risk cohorts at the community level.
- **Exclusion of Non-Claims-Based Interventions:** Programs or services that do not generate claims, such as community-based interventions, case management, or preventive outreach, may not be captured, even though they play a critical role in value-based care performance.
- **Selection Bias on Participation:** Providers or health plans that participate in value-based programs (e.g., ACOs) may differ systematically from those that do not, introducing selection bias. Even with statistical controls like propensity score matching, unobserved confounders could influence results.

These limitations highlight the importance of interpreting findings within the context of the data's scope and structure. Despite these constraints, the datasets still offer valuable insights into large-scale trends and systemic patterns that can inform policy and operational strategies for value-based care.

Ethical considerations

When conducting research using healthcare datasets, especially those involving patient demographics, clinical conditions, or payment information, it is critical to uphold ethical standards that protect individuals' rights and maintain data integrity. Here are the key ethical considerations:

Ethical Issue	Importance	Action
Data Privacy & Confidentiality	Patient information must be protected to ensure it is not traceable to individuals.	Use only de-identified or anonymized datasets (e.g., HIPAA-compliant), and avoid combining data that could lead to re-identification.
Informed Consent	Data subjects should ideally know how their information will be used.	Use publicly available datasets where informed consent is either already handled or not required (e.g., CMS datasets). For primary data collection, ensure informed consent is obtained.
Data Security	Prevent unauthorized access to sensitive data.	Store data securely, restrict access to authorized users, and follow institutional or governmental cybersecurity guidelines.
Ethical Use of Results	Misuse of research (e.g., to deny services or unfairly cut costs) can harm patients.	Ensure findings are used to improve healthcare quality, access, or efficiency, not to discriminate against or reduce care quality.
Regulatory Compliance	Research must follow laws like: HIPAA (U.S.), GDPR (EU, if applicable)	Ensure compliance with all applicable data governance rules.

By integrating financial, clinical, and demographic data across multiple dimensions of care delivery, this study seeks to offer a nuanced understanding of how value-based models function in real-world settings.

DATA ANALYSIS AND FINDINGS

Research Question- 1: *How does the number of providers affect the percentage of eligible patients receiving services across different counties and states?*

Data set used: Medicaid Managed Care

The correlation between the **number of healthcare providers** and the **percentage of eligible patients receiving services** offers an important lens to **healthcare accessibility and efficiency** across regions.

Higher Provider Availability → Higher Service Utilization

- States like Michigan and Arizona, with higher total provider counts (over 1.5 million combined), show broader coverage of eligible patients across most counties.
- Counties with larger provider bases, such as Wayne (MI), Kent (MI), and Maricopa (AZ), demonstrate higher service percentages, suggesting that greater availability of providers supports better utilization of services.

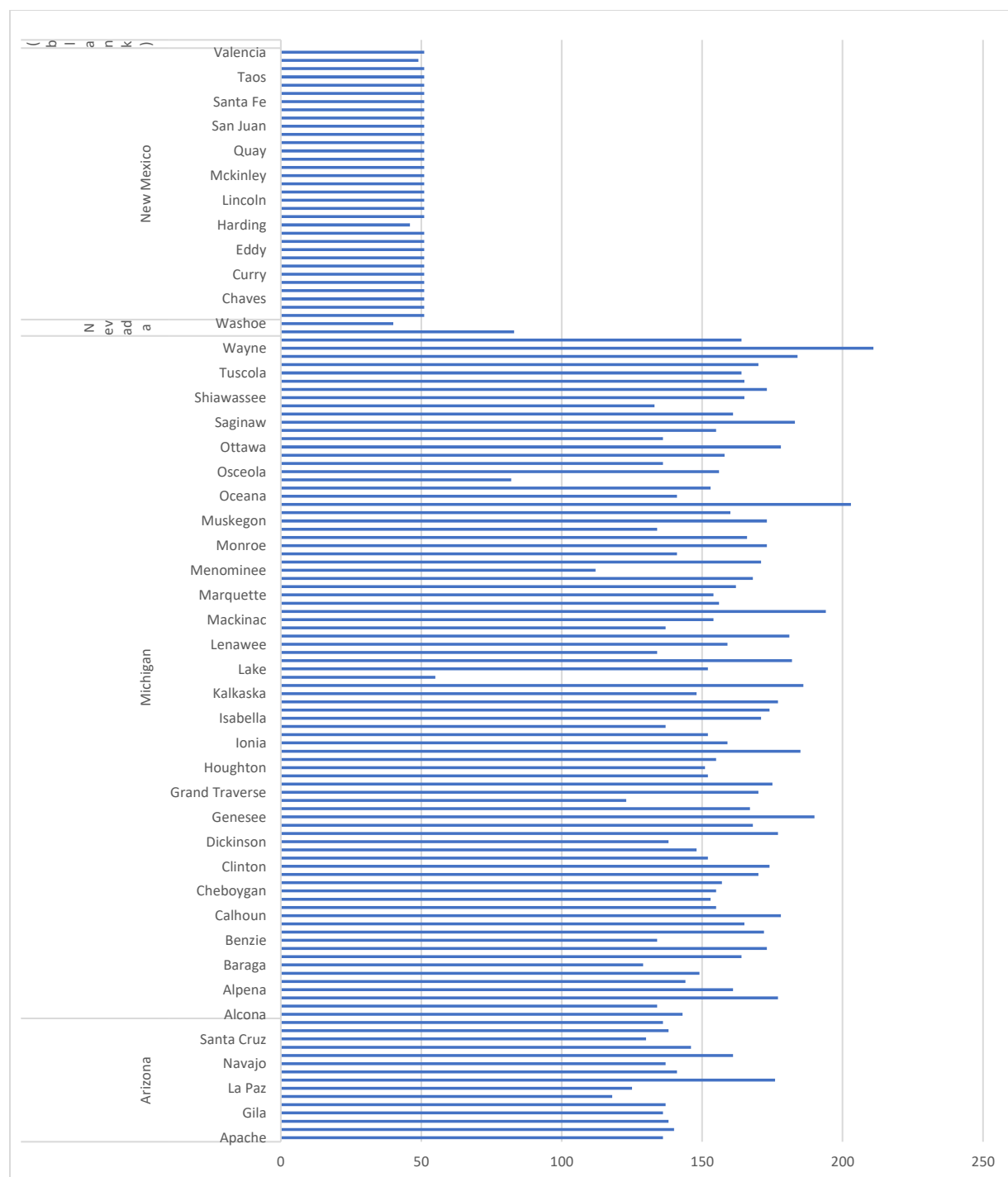
Low Provider Density Limits Access

- In New Mexico, despite pockets like Bernalillo showing moderate provider counts, many rural counties (e.g., Mora, De Baca) report low provider availability, often correlating with lower percentages of eligible patients receiving care.
- This trend reflects known national disparities where rural areas face provider shortages, leading to underserved populations despite eligibility.

Geographic Variation

- County-level differences within the same state (e.g., Apache vs. Maricopa in AZ) highlight micro-level disparities. High provider concentration in urban centers tends to fare better in service rates than rural areas.

Figure 1: Total healthcare providers vs percentage of eligible patients receiving services grouped by country and State



Research Question- 2: Which MCOs or plan categories deliver the highest number of services per active patient, and what does this suggest about care intensity or efficiency?

Data set used: Medicaid Managed Care

Comprehensive MCOs dominate both **patient count** and **service variety**, underscoring their central role in Medicaid Managed Care. Specialty plans (Behavioral, Dental) are narrower in scope but essential for targeted care delivery. This distribution highlights how plan type impacts access and specialization in patient services.

Comprehensive MCOs have the highest number of active patients (11,211), indicating that these plans serve the broadest population across multiple service categories:

- **Behavioral Health:** 5,079 patients
 - **Pediatric Dental:** 1,305 patients
 - **Prenatal Ob/Gyn:** 4,827 patients
- This reflects their comprehensive and integrated care delivery.

Behavioral Health Inpatient Specialty Plans (PIHP) show 4,030 active patients, all under Behavioral Health. This suggests these plans are highly specialized and focus solely on inpatient mental health services.

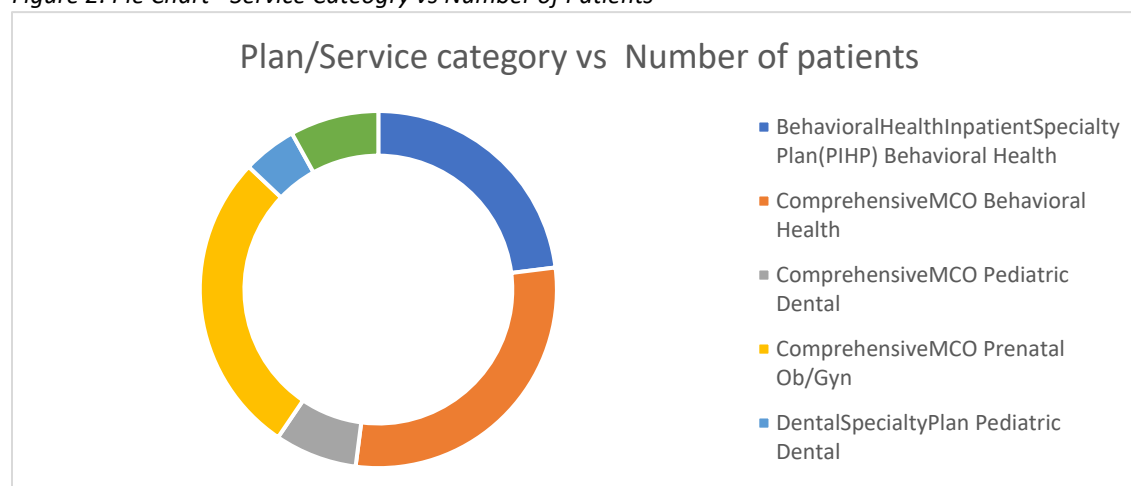
Outpatient Specialty Health Plans (PAHP) account for 1,408 patients, also entirely in Behavioral Health. These are likely to provide ambulatory or community-based mental health services.

Dental Specialty Plans have 855 active patients, all in Pediatric Dental, indicating a targeted dental care offering for children.

Tabular view - Service Category vs Number of Patients

Plan Category —> Service category	Count of Number of Active Patients
Behavioral Health Inpatient Specialty Plan(PIHP)	4030
Behavioral Health	4030
Comprehensive MCO	11211
Behavioral Health	5079
Pediatric Dental	1305
Prenatal Ob/Gyn	4827
Dental Specialty Plan	855
Pediatric Dental	855
Outpatient Specialty Health Plan(PAHP)	1408
Behavioral Health	1408
Grand Total	17504

Figure 2: Pie Chart - Service Category vs Number of Patients



Research Question- 3: What are the trends in the number of active patients and the number of services per active patient across states?

Data set used: Medicaid Managed Care

Arizona:

- Total Active Patients: **~5.9 million** & Total Services per Active Patient: **~46,163**
- Shows a **moderate growth trend** in both active patients and services per patient. The rise from **8,647 (2020)** to **9,623 (2024)** services per patient indicates a slight **increased in care intensity** over time.

Michigan:

- Total Active Patients: **~6.1 million** (highest among the states) & Services per Active Patient: **~141,859** (also highest)
- Shows the **strongest care intensity** growth, from **24,921 (2020)** to **38,301 (2024)**, suggesting significant increases in patient engagement, healthcare utilization, or possibly more comprehensive service delivery.

Nevada:

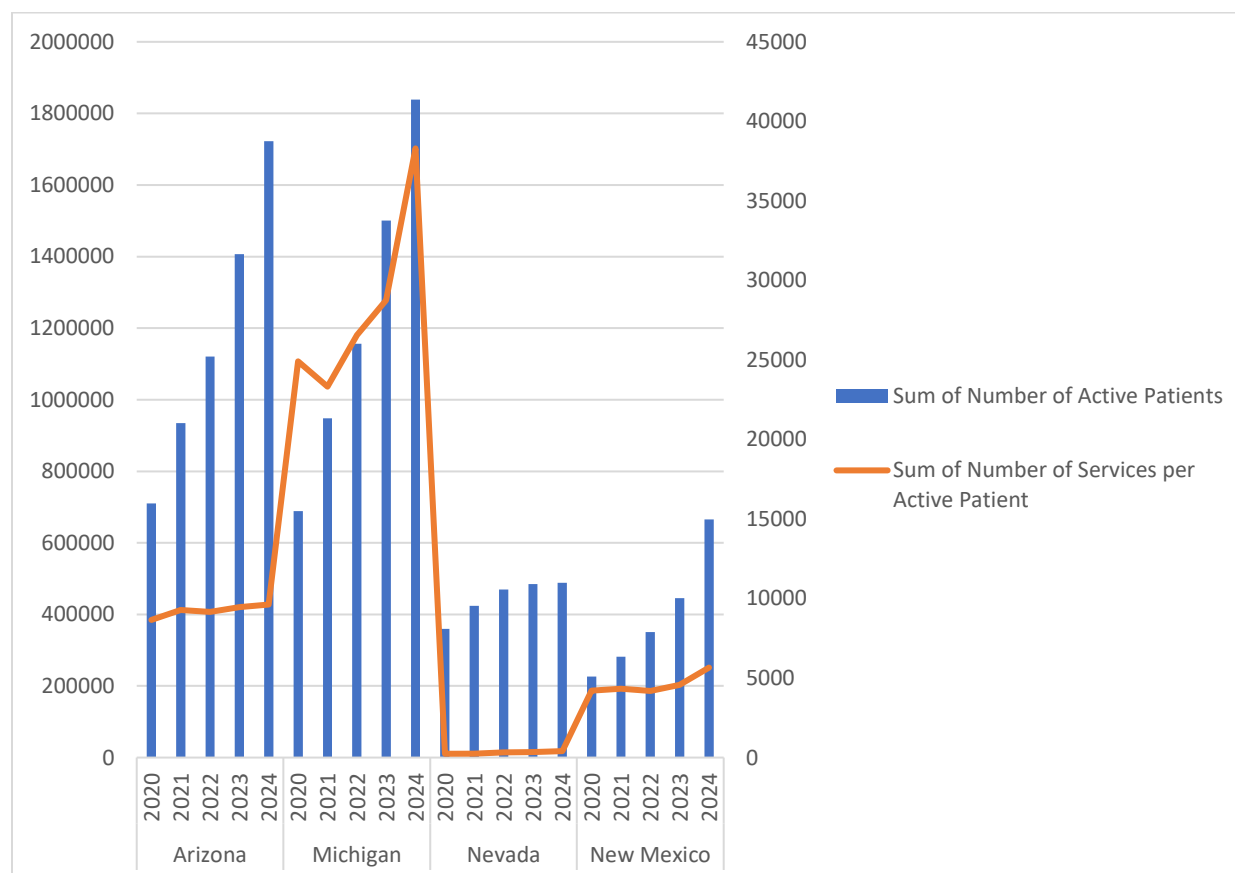
- Total Active Patients: **~2.2 million** & Services per Active Patient: **1,554** (lowest overall)
- Care intensity is **relatively low** and only slightly increasing year over year. This may reflect **access issues, lower provider availability, or more limited benefit structures**.

New Mexico:

- Total Active Patients: **~2 million** & Services per Active Patient: **~22,938**
- Shows **steady service growth**, rising from **4,203 (2020)** to **5,651 (2024)**. This reflects **moderate improvements** in service delivery or access expansion.

Michigan leads in both the number of active patients and care intensity, indicating a robust service delivery infrastructure. Nevada lags significantly behind, which may require targeted policy action or healthcare investment to increase services. All states are improving, but the rate of change varies, pointing to regional disparities in care access or demand that may warrant deeper analysis.

Figure 3: Bar-Line Graph – count of Active Patients vs Number of Services grouped by State



Research Question- 4: How do average Medicare payments compare to average submitted charges across different DRGs?

Data set used: Medicare IP Hospitals by Geography and Service

Highest Payments and Charges:

- Internal Major Procedures has the highest Sum of Avg_Mdcr_Pymt_Amt of \$120,645,588.7 and the highest Sum of Avg_Submtd_Cvrd_Chrg of \$660,861,170.8.
- Other high-value categories are Cardio and Pulmonary Related, Other Complications, and Spine and Bone, which have substantial values for both metrics.

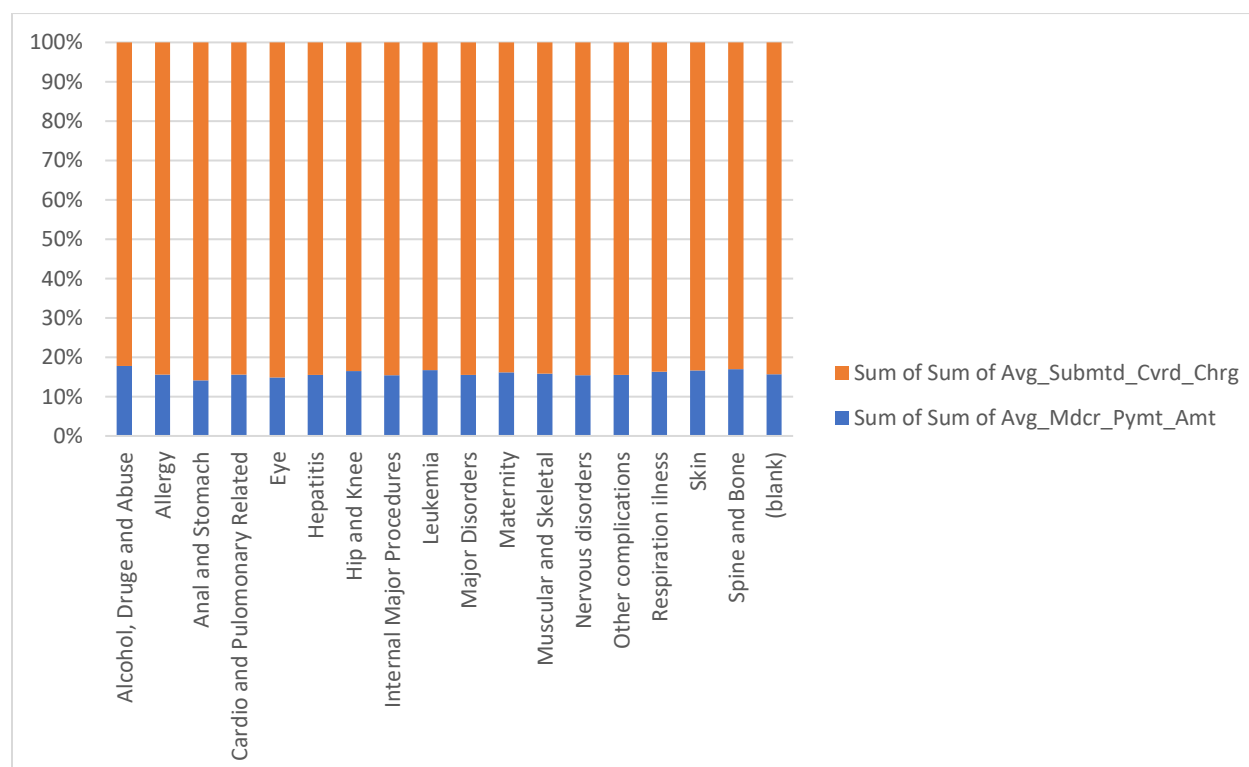
Category Analysis:

- Cardio and Pulmonary Related stands out for having a significantly higher ratio of charges to Medicare payments, with \$99,987,909.38 in payments and \$542,834,544.2 in charges. This suggests that for this category, the payments are relatively lower compared to the billed charges, which may indicate issues with coverage or reimbursement rates.
- Anal and Stomach also has a high Sum of Avg_Submtd_Cvrd_Chrg (\$50,735,449.23) compared to its Medicare payments (\$8,323,107.68), indicating a similar discrepancy.
- Alcohol, Drug and Abuse has a low Sum of Avg_Mdcr_Pymt_Amt (\$1,526,438.45) compared to its Sum of Avg_Submtd_Cvrd_Chrg (\$7,037,082.67), suggesting that the payments are not as high despite substantial charges.

High Variability:

- **"Blank"** appears to be an outlier with a very large total: **\$485,930,469.9** in Medicare payments and **\$2,613,806,761** in covered charges. It likely represents aggregated data from multiple categories
- **Categories with Significant Payment Disparities:** Categories like **Cardio and Pulmonary Related**, **Internal Major Procedures**, and **Other Complications** display some of the highest charge-to-payment ratios, which may warrant further investigation into how payments are calculated or whether there is a misalignment between billed charges and actual payments.

Figure 4: Stacked Bar Chart – Sum of Total Submitted Claim amount vs Total Medicare Paid Amount – grouped by DRG category



Research Question- 5: Which DRGs have the highest total discharges, and what are the corresponding average payment amounts and submitted charges?

Data set used: Medicare IP Hospitals by Geography and Service

An analysis of DRG categories by total discharge volume reveals that **"Other complications"** ranks the highest with over 1.1 million discharges, followed by **"Cardio and Pulmonary Related"** at approximately 642,000, and **"Respiration Illness"** with around 337,000.

These three categories represent a substantial portion of inpatient volume, indicating areas of high clinical and operational demand. **"Hip and Knee"** procedures and **"Maternity"** services also appear in the top five, with discharge volumes of over 154,000 and 147,000, respectively.

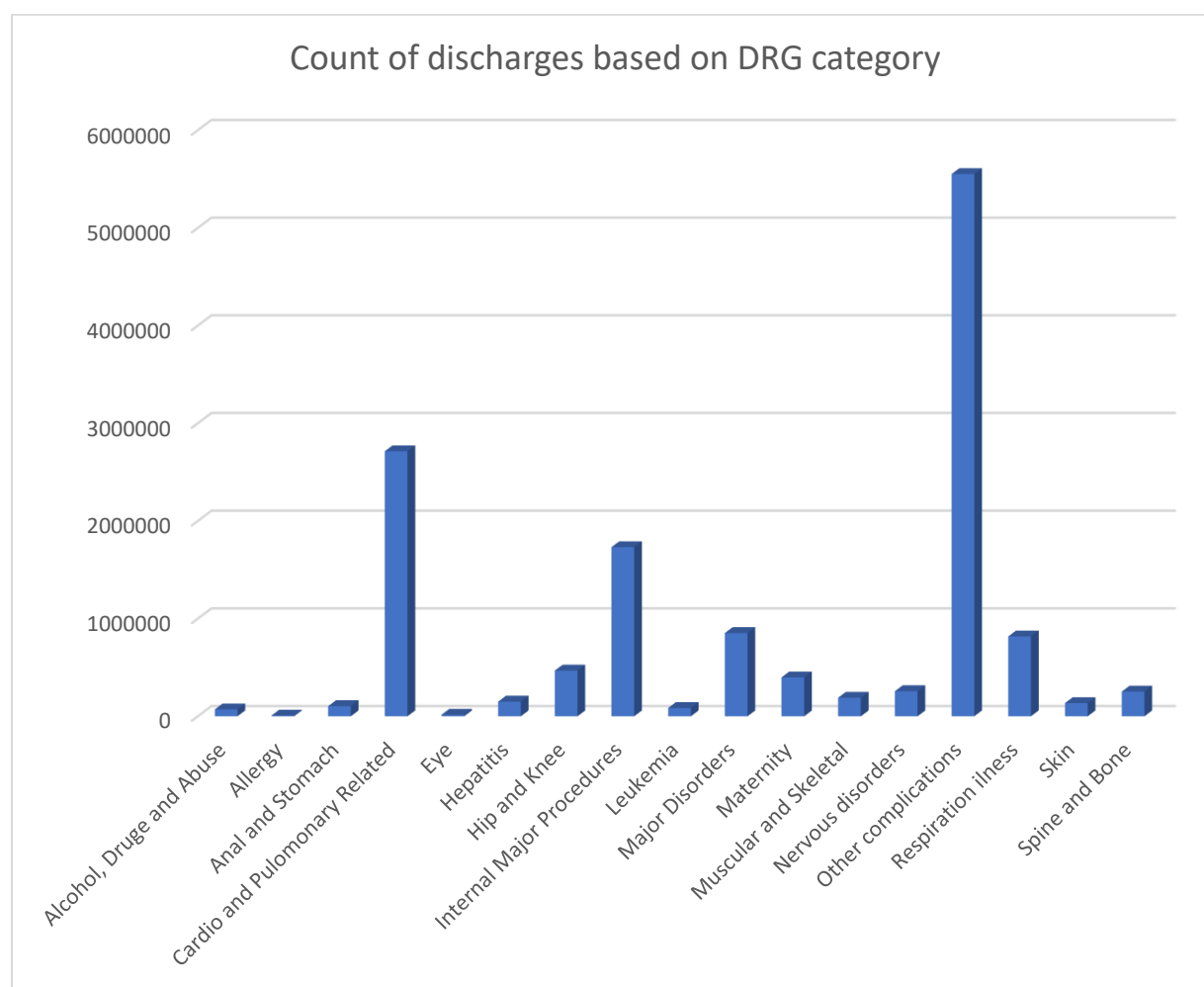
When paired with financial data, **Cardio and Pulmonary** emerge as the most resource-intensive category, with average Medicare payments totaling over **\$99 million** and submitted charges exceeding **\$542 million**, pointing to a significant financial footprint.

In contrast, while **Other Complications** leads in volume, it accounts for around **\$14 million** in average Medicare payments and nearly **\$78 million** in submitted charges, suggesting a broader but less cost-intensive case mix. **Hip and Knee**, and **Maternity** services also represent high-spend areas, supporting their relevance for a value-based care focus.

Tabular view of DRG categories against key metrics:

DRG Categories	Max of the Sum of Tot_Dschrsgs	Min of Sum of Tot_Dschrsgs	Average of the Sum of Tot_Dschrsgs	Count of DRG groups
Alcohol, Drugs, and Abuse	38898	5542	17634	4
Allergy	4175	3435	3805	2
Anal and Stomach	35840	107	4333.666667	24
Cardio and Pulmonary Related	642212	28	29491.5	92
Eye	6377	143	3104	5
Hepatitis	33188	74	10011.46667	15
Hip and Knee	154166	420	35991.53846	13
Internal Major Procedures	164552	15	9309.55914	186
Leukemia	16638	158	4711.055556	18
Major Disorders	182122	29	20788.39024	41
Maternity	147308	14	11377.05714	35
Muscular and Skeletal	27880	667	9506.5	20
Nervous disorders	52930	335	13538.73684	19
Other complications	1125792	11	25118.80995	221
Respiration illness	337938	384	54524.2	15
Skin	25353	77	4658.448276	29
Spine and Bone	61274	26	15835.0625	16

Figure 5: Bar Chart (3D) - DRG Category vs Discharge counts



Research Question- 6: How do average Medicare payments compare to submitted charges across different DRGs and provider locations (urban vs. rural)?

Data set used: Medicare IP Hospitals by Provider and Service

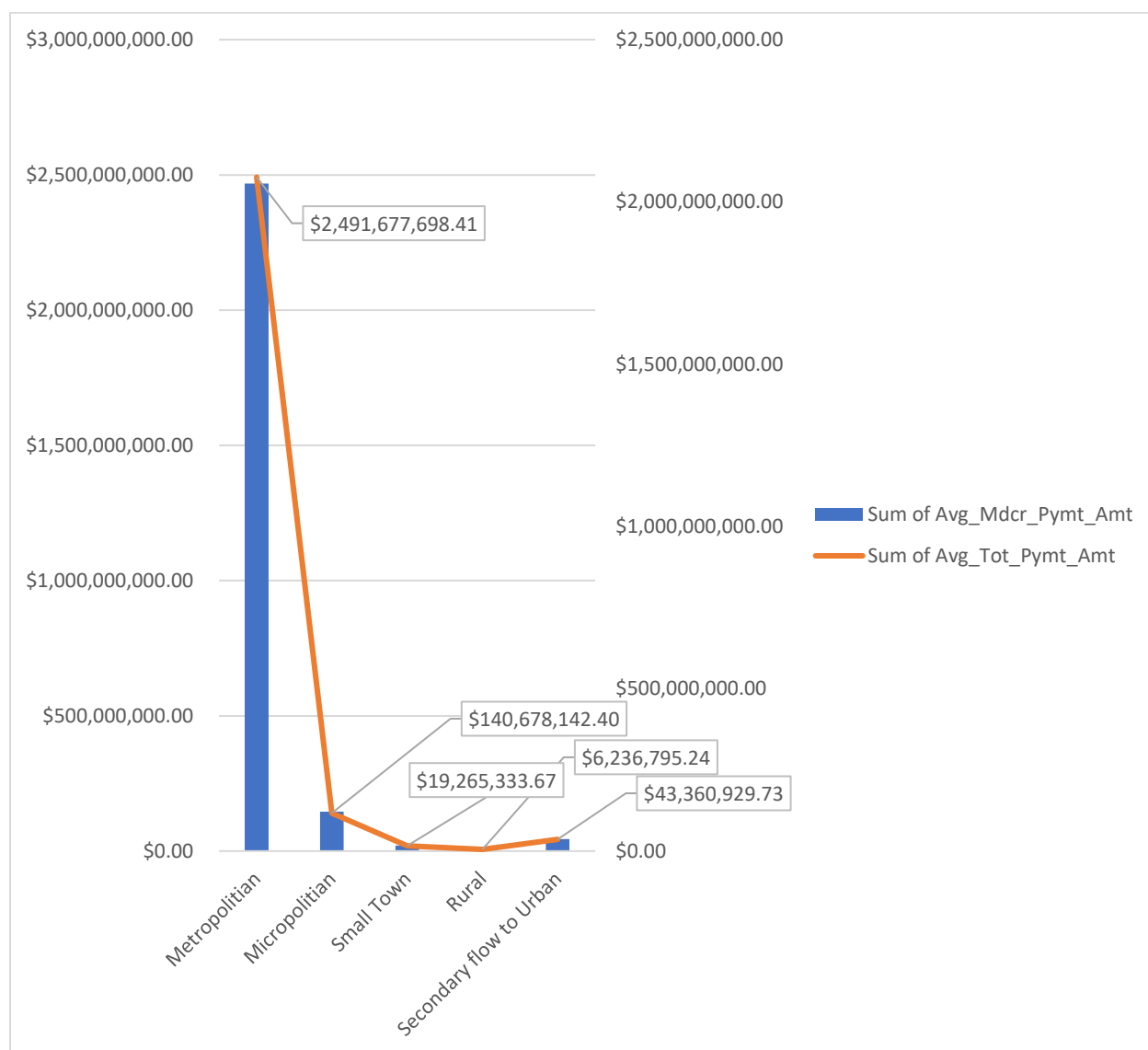
Medicare Payments vs. Submitted Charges (Across DRGs)

- Across nearly all DRG categories, submitted charges significantly exceed average Medicare payments
 - Cardio and Pulmonary Related DRGs: ~\$99.9M paid vs. ~\$542.8M submitted.
 - Hip and Knee: ~\$10.05M paid vs. ~\$51M submitted.
 - The payment-to-charge ratio typically ranges from 15% to 25%, meaning Medicare pays only a fraction of what is billed.
- DRGs with high procedural complexity or specialized care tend to have greater gaps between submitted charges and payments, especially where provider markups are higher.

Urban vs. Rural Payment Dynamics

- Metropolitan areas had the highest Medicare payments (\$2.49B) and submitted charges.
- Rural areas showed dramatically lower values (~\$6.2M), despite potentially serving large regions geographically.
- Micropolitan and Small Towns also lag far behind in both metrics.
- Urban providers handle more high-cost, high-volume cases, resulting in larger payments and charges.
- Rural and small-town providers face lower service volumes and possibly lower reimbursement rates, highlighting access disparities.

Figure 6: Bar Line Graph – Sum of Average Medicare Paid amount and Total Paid amount against MSA category



OUTCOME OF HYPOTHESES

Hypotheses Description	Outcome	Reason
Evaluate if certain plan types (e.g., HMOs vs. PPOs) deliver more services per patient, which may reflect care intensity or access.	Null Hypothesis - Rejected	Plan types do differ in how intensively they deliver services, reflecting variations in care intensity or access.
Assess discrepancies between billed and reimbursed amounts to identify inefficiencies or payment gaps.	Null Hypothesis - Rejected	There is a significant gap between what is billed and what is paid, indicating payment inefficiencies or policy-driven reimbursement limits.
MCOs or plan categories that deliver a higher number of services per active patient are associated with increased care intensity, but not necessarily higher care efficiency.	Null Hypothesis - Rejected	Increased services per patient do not consistently reflect higher efficiency, supporting the alternate hypothesis.
Understand care access and service utilization in different geographic settings.	Null Hypothesis - Rejected	There is a significant difference in service utilization by geography, indicating access or capacity gaps between urban and rural areas.

CONCLUSION

This capstone project demonstrates that value-based care (VBC) models can improve care quality and efficiency but may also unintentionally widen healthcare disparities if financial incentives and provider distribution are not aligned with patient needs. Analyzing large-scale claims data revealed that states with higher provider density, like Michigan, achieve better patient coverage and higher care intensity, while rural areas face persistent access gaps. Significant discrepancies between Medicare payments and submitted charges, along with urban-rural differences in service volumes, underscore the need for targeted strategies to ensure equitable care delivery.

These findings highlight that refining VBC programs to address regional and demographic disparities is essential for building a sustainable, fair healthcare system that delivers quality outcomes for all patients, especially high-risk and underserved populations.

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